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AVA-VLA: Improving Vision-Language-Action models with Active Visual Attention

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Abstract

Vision-Language-Action (VLA) models have shown remarkable progress in embodied tasks recently, but most methods process visual observations independently at each timestep. This history-agnostic design treats robot manipulation as a Markov Decision Process, even though real-world robotic control is inherently partially observable and requires reasoning over past interactions. To address this mismatch, we reformulate VLA policy learning from a Partially Observable Markov Decision Process perspective and propose AVA-VLA, a framework that conditions action generation on a recurrent state that serves as a neural approximation to the agent's belief over task history. Built on this recurrent state, we introduce Active Visual Attention (AVA), which dynamically reweights visual tokens in the current observation to focus on regions most relevant given both the instruction and execution history. Extensive experiments show that AVA-VLA achieves state-of-the-art performance on standard robotic benchmarks, including LIBERO and CALVIN, and transfers effectively to real-world dual-arm manipulation tasks. These results demonstrate the effectiveness of temporally grounded active visual processing for improving VLA performance in robotic sequential decision-making. The project page is available at https://liauto-dsr.github.io/AVA-VLA-Page.

1. Introduction

Recent advances in robotic manipulation have demonstrated impressive progress in training robot action policies that can act across diverse real-world tasks. One transformative paradigm is Vision-Language-Action (VLA) models [2–4, 7, 19, 22, 36, 65], which integrate visual perception, natural language understanding, and action generation within a unified neural architecture. These models, which are capable of instruction following and robotic action generation, exhibit strong understanding and generalization abilities after being fine-tuned for downstream scenarios.

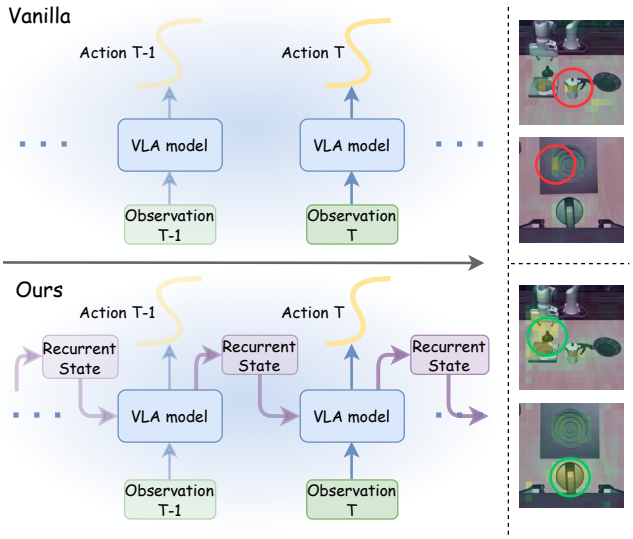


Figure 1. (a) Visualized comparison of the proposed AVA-VLA framework and vanilla VLAs. (b) Qualitative comparison of visual focus from two viewpoints while executing the task "turn on the stove and put the moka pot on it." The vanilla OpenVLA-OFT [20] baseline fails to locate the task-critical "stove" switch, whereas AVA-VLA exhibits more stable focus by leveraging historical context.

To adopt the ability to understand diverse scenes, objects, and language instructions, most VLA models are built upon pretrained Vision-Language Models (VLMs) [9, 18, 29]. Such models typically extend VLM architectures with modules such as action tokenization [19, 45] or specialized action experts [22, 57] to enable action-oriented outputs. Based on this architectural inheritance, these VLA models typically process visual inputs as isolated temporal frames, treating each frame independently. This implicitly formulates robot manipulation as a Markov Decision Process (MDP) [16, 31], where actions are generated from the current visual observation, assumed to represent the complete world state. In realistic robotic manipulation, how-

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ever, the current visual frame is only a partial observation of the environment state. This full state includes unobservable dynamics across time, such as internal states and occluded information. By discarding the rich context from the past, this MDP-based approach is suboptimal for the dynamic sequential decision-making required in robotic manipulation.

This limitation of the MDP-based assumption has significant impacts on VLA models, particularly for the model's visual processing capabilities. VLA modeling is essentially a dynamic feedback control problem, where each preceding action directly alters the current visual input. However, by processing frames in isolation, the visual attention weights, guided by the static language instruction, are forced to re-evaluate the independent visual information from scratch at each decision step. Without global context, the model cannot effectively suppress temporally redundant information and focus on regions made important by past actions. As a result, the visual system remains passive rather than active. In fact, the inability to anticipate perceptual intent a priori makes active visual modules difficult to realize in computer vision. However, the sequential dynamics of decision-making create an opportunity for active visual perception. Recognizing the limitations of processing frames in isolation, some recent methods [24, 52, 58] have begun to leverage historical information, such as frame comparison results and KV-cache reuse, for efficient visual token processing [8, 62]. However, these approaches mainly focus on visual token pruning for efficiency. Therefore, designing a dynamic, context-aware visual processing paradigm that improves both visual processing and VLA generalization remains a significant challenge.

To address this challenge, we propose AVA-VLA, inspired by the Partially Observable Markov Decision Process (POMDP) framework [21, 44]. We observe that the core challenge identified above is similar to the POMDP challenge of forming a robust belief state, which functions as a summary of past observations and actions to guide decision-making under uncertainty. Since directly computing or representing the belief state is generally intractable, we introduce a recurrent state, which functions as a neural approximation of this belief state and is calculated by the intermediate output of the model in the previous time step. Then, we design an Active Visual Attention (AVA) module to leverage this recurrent state to calculate the importance of visual tokens and dynamically modulating the visual processing of the current frame. This allows the model to filter and focus its attention based on its historical belief, rather than purely static language instruction. Therefore, the proposed AVA-VLA framework does not rely solely on the current observation but learns to explicitly condition the action prediction on the recurrent state.

Through extensive experiments in both simulation benchmarks [11, 28, 31] and real-world tasks [5], we demonstrate that our proposed active visual attention module helps improve policy performance compared to previous VLA frameworks. Our contributions are threefold:

- We propose the novel AVA-VLA framework to solve the critical limitation of lacking historical context in MDP-based VLA models. To our knowledge, it is the first VLA framework to explicitly address this limitation via a POMDP-inspired approach.
- We introduce an Active Visual Attention (AVA) module that leverages the recurrent state to dynamically modulate the visual processing of the current frame for action prediction.
- We conduct comprehensive evaluations in both simulation and real-world tasks, demonstrating that the AVA-VLA framework improves VLA performance, and our method achieves state-of-the-art performance across multiple robot tasks.

2. Related Work

Vision-Language-Action Models. VLMs [9, 18, 29, 42] have been pivotal in advancing robotic control by providing rich multi-modal representations. This has fostered the development of VLA models [2–4, 7, 19, 22, 36, 37, 46, 65] that bridge high-level perception with low-level action generation. A significant paradigm shift was the introduction of action tokenization by the RT series [1, 4, 65]. This approach treats control as a sequence modeling problem, enabling scalable web-to-robot transfer. Models like OpenVLA [19] and UniVLA [5] generate action policies in the autoregressive (AR) manner. While expressive, the sequential nature of AR decoding is computationally intensive. Therefore, recent research has diversified into more efficient and effective action decoding strategies. Models such as CogACT [22] and the π series [3, 36], have explored diffusion-based decoders for the iterative refinement of continuous action trajectories. Other recent works, such as OpenVLA-OFT [20] and its variant [23], employ parallel decoding, which enables the simultaneous prediction of actions within the action chunk, improving inference efficiency and supporting scalable deployment.

Sequential Processing in VLMs. Many VLM studies [10, 33, 38, 53] focus on processing sequential visual data for tasks such as video understanding and temporal-based video questions. These works efficiently aggregate historical information, allowing the model to build a holistic, temporal-aware representation of the video's content. VLM-3R [10] employs a geometry encoder to derive implicit 3D tokens that represent spatial-temporal understanding. [53] incrementally updates a persistent internal state that encodes the scene content. These VLM models use history for passive comprehension or offline understanding. In contrast, VLA models operate in an active, dynamic decision-making environment, which requires the model to

interact with the environment. This distinction motivates our POMDP-inspired approach, which focuses on maintaining a recurrent state for active decision-making.

3. Methods

In this section, we present our proposed VLA method. We begin with the preliminaries (3.1), followed by the AVA-VLA framework (3.2) and the detailed description of the proposed Active Visual Attention module (3.3). We then explain our training and inference procedures (3.4). An overview of the proposed AVA-VLA framework is shown in Figure 2.

3.1. Preliminaries

A typical VLA model $\mathcal{P}_\theta$, parameterized by θ, consists of four main components: a Large-Language-Model (LLM) backbone $\mathcal{M}$, a vision encoder $\mathcal{E}$, a language tokenizer $\mathcal{T}$, and an action head (or de-tokenizer) $\mathcal{Q}$. We thus define the model as $\mathcal{P} = \{\mathcal{M}, \mathcal{E}, \mathcal{T}, \mathcal{Q}\}$.

Following the representative OpenVLA [19], at timestep t, given an input tuple $\boldsymbol{x}^t = (\boldsymbol{x}_I^t, \boldsymbol{x}_S^t)$, the visual encoder $\mathcal{E}$ first encodes the input image $\boldsymbol{x}_I^t$ into L_I visual tokens: $\boldsymbol{z}_I^t = \mathcal{E}(\boldsymbol{x}_I^t) \in \mathbb{R}^{\mathrm{L}_I \times d}$, where d denotes the embedding dimension. These visual tokens are then concatenated with L_S^t language tokens, $\boldsymbol{z}_S^t = \mathcal{T}(\boldsymbol{x}_S^t) \in \mathbb{R}^{\mathrm{L}_S^t \times d}$. The combined sequence is then fed into the LLM backbone $\mathcal{M}$ to generate output hidden states $\boldsymbol{h}^t$. Finally, the action head $\mathcal{Q}$ maps the output hidden states $\boldsymbol{h}^t$ into a D-dimensional executable action $\mathcal{A}^t$ for robotic control (e.g., $D = 7$ for 3-DoF translation, 3-DoF rotation, and binary gripper control). Thus, the entire forward pass at timestep t can be formulated as:

$$\mathcal{A}^t = \mathcal{Q}(\boldsymbol{h}^t) = \mathcal{Q}(\mathcal{M}(\boldsymbol{z}_I^t, \boldsymbol{z}_S^t)). \tag{1}$$

Recent representative VLA models, such as OpenVLA-OFT [20] and its variant [23], map the output hidden states into an executable action chunk $\mathcal{A}^t = [a_0^t, a_1^t, ..., a_{\mathrm{L}_c-1}^t] \in \mathbb{R}^{\mathrm{L}_c \times D}$, where L_c and D represent the length of the action chunk and the dimensionality of each atomic action, respectively. To facilitate parallel generation, a learnable action placeholder embedding $\boldsymbol{p}^t$ is appended to the input sequence [23, 27]. This placeholder embedding is set to empty in OpenVLA-OFT, i.e., $\boldsymbol{p}^t = \bar{\mathbf{0}} = [\mathbf{0}_0, \mathbf{0}_1, ..., \mathbf{0}_{\mathrm{L}_c-1}] \in \mathbb{R}^{\mathrm{L}_c \times D \times d}$. The corresponding forward pass under parallel decoding at timestep t can thus be expressed as:

$$\mathcal{A}^t = \mathcal{Q}(\mathcal{M}_{\text{parallel}}(\boldsymbol{z}_I^t, \boldsymbol{z}_S^t, \boldsymbol{p}^t)). \tag{2}$$

Regardless of whether AR or parallel decoding is used, these VLA models learn to predict the action $\bar{\mathcal{A}}^t$ only from the current observation $\boldsymbol{x}^t$. This implicitly models the task as a Markov decision process:

$$\bar{\mathcal{A}}^t \sim \mathcal{P}_\theta(\mathcal{A}^t \mid \boldsymbol{x}^t). \tag{3}$$

3.2. AVA-VLA Framework

The history-agnostic design for policy learning in Eq. (3) is suboptimal for effective visual token processing in dynamic sequential decision-making [21, 61], as it fails to capture non-observable dynamics or occluded information. This limitation inspired us to re-formulate the VLA model from a POMDP perspective. In a POMDP framework, the optimal policy at timestep t should be conditioned not only on the current observation $\boldsymbol{x}^t$ but also on a belief state b^{t-1}, which captures all relevant historical context, including observations and actions, i.e., $b^{t-1} = P(s_{t-1} \mid \boldsymbol{x}^{<t}, \mathcal{A}^{<t})$. Inspired by this theoretical framework, we re-formulate the VLA policy as

$$\bar{\mathcal{A}}^t \sim \mathcal{P}_\theta(\mathcal{A}^t \mid \boldsymbol{x}^t, b^{t-1}). \tag{4}$$

This formulation provides a theoretical foundation for designing a more effective visual processing paradigm, suggesting that leveraging historical context in observations can improve VLA generalization. Since computing the theoretical belief state b^{t-1} is generally intractable, we instead propose to learn a compressed representation, $\boldsymbol{r}^{t-1}$, as its neural approximation. This approach naturally transforms the VLA model into a recurrent structure [30], leading to a non-Markovian policy conditioned on this learned representation: $\bar{\mathcal{A}}^t \sim \mathcal{P}_\theta(\mathcal{A}^t \mid \boldsymbol{x}^t, \boldsymbol{r}^{t-1})$.

In our proposed AVA-VLA framework, we term this approximate vector $\boldsymbol{r}^{t-1}$ the recurrent state, which captures historical context. In typical VLA models, the hidden states immediately preceding action generation contain fused visual and language information and are predictive of the agent's intent. Therefore, we derive the recurrent state for timestep t from the action-related hidden state at timestep $t-1$.

Specifically, for a parallel-decoding-based VLA model, which contains M decoder layers that predict $\mathrm{L}_A = \mathrm{L}_c D$ actions in one forward pass, we denote its hidden states output at the m-th layer and time t by $h_m^t \in \mathbb{R}^{\mathrm{L}_A \times d}$. The corresponding recurrent state is computed by:

$$\boldsymbol{r}^{t-1} = \mathcal{B}(\boldsymbol{h}_M^{t-1}) \in \mathbb{R}^{\mathrm{L}_A \times d}, \tag{5}$$

where $\mathcal{B}$ is an MLP module that transforms the hidden state into the recurrent state.

We employ this recurrent state to guide the VLA model to actively focus on visual regions that become critical along the time sequence. To utilize the recurrent state, we introduce the active visual attention module by quantifying the importance of visual tokens and dynamically modulating the processing of the visual frame for the current timestep.

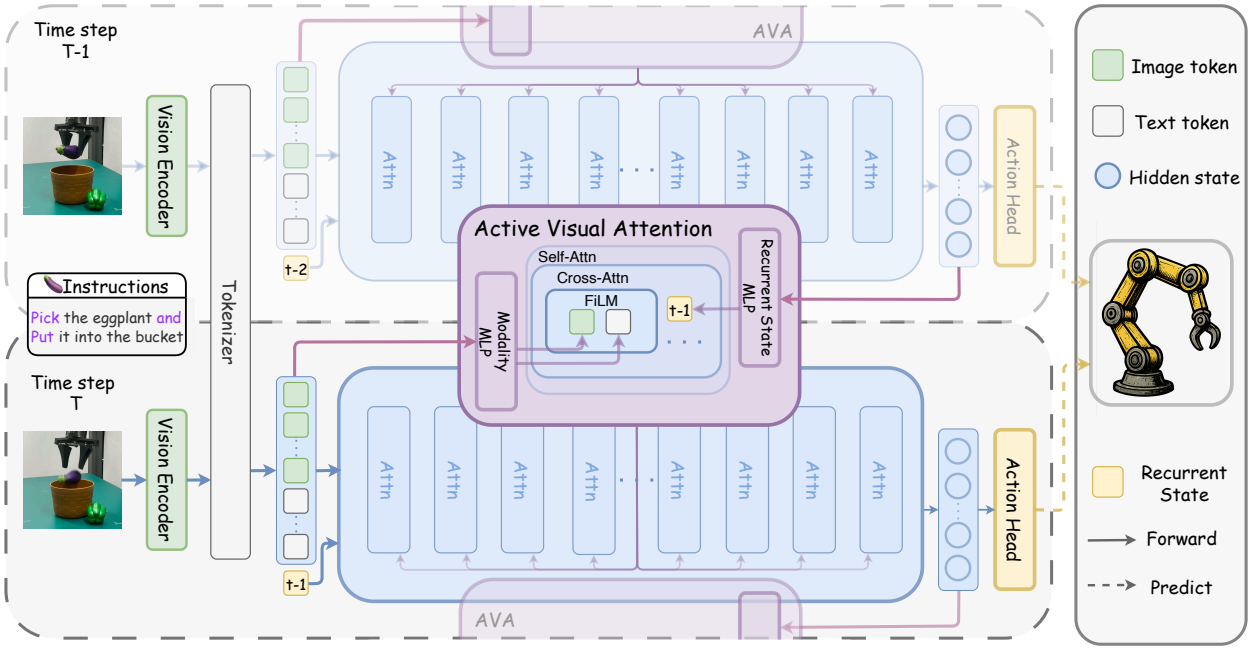


Figure 2. Overview of the proposed AVA-VLA framework. At each timestep, the recurrent state is projected from the previous hidden state to preserve historical context and to initialize the current action tokens. Then the AVA module combines this recurrent state with text-conditioned visual features from the current observation to generate soft importance scores, which modulate the visual attention matrices throughout the backbone LLM, enabling the model to focus on task-relevant regions based on both temporal context and current perception.

Moreover, in order to preserve the rich historical information, we use this recurrent state $\boldsymbol{r}^{t-1}$ for action placeholder [20, 23] embedding initialization, i.e., $\boldsymbol{p}^t = \boldsymbol{r}^{t-1}$.

For simplicity, our framework is built upon the OpenVLA-OFT foundation model. Therefore, the forward pass at timestep t, incorporating the AVA module and state-based initialization, is formulated as:

$$\mathcal{A}^t = \mathcal{Q}(\mathcal{M}_{\text{parallel}}(\boldsymbol{z}_I^t, \mathcal{V}(\boldsymbol{x}^t, \boldsymbol{r}^{t-1}), \boldsymbol{z}_S^t, \boldsymbol{r}^{t-1})), \quad (6)$$

where $\mathcal{V}$ is the proposed AVA module, which takes the current observations and the recurrent state as input.

3.3. Active Visual Attention

We now describe the detailed architecture of the AVA module $\mathcal{V}$, which is designed to modulate visual processing in a dynamic manner.

The AVA module first employs modality-specific MLPs to encode the visual features $\boldsymbol{z}_I^t$ and the instruction feature $\boldsymbol{z}_S^t$ into $\bar{\boldsymbol{z}}_I^t \in \mathbb{R}^{\mathrm{L}_I \times d'}$ and $\bar{\boldsymbol{z}}_S^t \in \mathbb{R}^{\mathrm{L}_S^t \times d'}$, respectively, where $d' < d$. A feature-wise linear modulation (FiLM) [35] is applied to condition the visual features on the language instruction, i.e., $\hat{\boldsymbol{z}}_I^t = \mathcal{F}_\gamma(\bar{\boldsymbol{z}}_S^t) \odot \bar{\boldsymbol{z}}_I^t + \mathcal{F}_\beta(\bar{\boldsymbol{z}}_S^t)$. Then it uses the visual tokens $\hat{\boldsymbol{z}}_I^t$ as the query,

$$\mathbf{Q}^t = W_Q \hat{\boldsymbol{z}}_I^t \in \mathbb{R}^{\mathrm{L}_I \times d'}, \quad (7)$$

and the recurrent state as the key and value

$$\mathbf{K}^t, \mathbf{V}^t = (W_K / W_V)\hat{\boldsymbol{r}}^{t-1} \in \mathbb{R}^{\mathrm{L}_A \times d'}, \quad (8)$$

where W_Q, W_K, W_V are linear projection layers, and $\hat{\boldsymbol{r}}^{t-1} \in \mathbb{R}^{\mathrm{L}_A \times d'}$ is the output of $\boldsymbol{r}^{t-1}$ after MLP encoding. Then it computes the attention matrix and feeds the output into a self-attention layer

$$\mathbf{O}^t = \text{Self-Att}\left(\text{Cross-Att}(\mathbf{Q}^t, \mathbf{K}^t, \mathbf{V}^t)\right). \quad (9)$$

Inspired by [40, 47], we feed the resulting tokens $\mathbf{O}^t \in \mathbb{R}^{\mathrm{L}_I \times d'}$ into a Feedforward Neural Network (FFN), a linear layer $\mathcal{W} : \mathbb{R}^{d'} \to \mathbb{R}^2$, and apply a Softmax function along the feature dimension. This predicts the logits for enhancing or weakening each visual token as:

$$\boldsymbol{\rho}^t = \text{Softmax}\left(\mathcal{W}\left(\text{FFN}\left(\mathbf{O}^t\right)\right)\right) \in \mathbb{R}^{\mathrm{L}_I \times 2}. \quad (10)$$

Then we compute the final soft weights for visual tokens at time t by $\boldsymbol{\omega}^t = \boldsymbol{\rho}^t \boldsymbol{\gamma}$, where $\boldsymbol{\gamma}$ is a 2-dimensional vector. These soft weights directly represent the importance scores of the visual tokens. The components of $\boldsymbol{\gamma}$, γ_0 and γ_1, represent the scalar scores for enhancing and weakening a visual token, respectively.

The soft weights vector $\boldsymbol{\omega}^t$ is applied to all layers of the LLM backbone. Specifically, at time step t, let the total sequence length be L_o^t, we denote the attention score of the m-th layer by $\mathbf{C}^{t,m} \in \mathbb{R}^{\mathrm{L}_o^t \times \mathrm{L}_o^t}$, which is calculated by applying the original attention mask to the raw attention scores. The final attention matrix of the m-th layer $\mathbf{A}^{t,m}$ is calculated by applying the soft attention mask matrix $\mathbf{U}^t$ with the

Softmax operation to $\mathbf{C}^{t,m}$:

$$\mathbf{A}_{i,j}^{t,m} = \frac{\exp(\mathbf{C}_{i,j}^{t,m})\mathbf{U}_{i,j}^{t}}{\sum_{l=1}^{\mathrm{L}_o^t}\exp(\mathbf{C}_{i,l}^{t,m})\mathbf{U}_{i,l}^{t}},\ 1 \le i,j \le \mathrm{L}_o^t, \quad (11)$$

where the soft attention matrix $\mathbf{U}^t$ is constructed based on the soft weights vector $\boldsymbol{\omega}^t$:

$$\mathbf{U}_{i,j}^{t} = \begin{cases} 1 & \text{if } i=j \text{ or } j \notin \Lambda_I, \quad 1 \le i,j \le \mathrm{L}_o^t \\ \boldsymbol{\omega}_j^t & \text{if } i \ne j \text{ and } j \in \Lambda_I,\ 1 \le i,j \le \mathrm{L}_o^t \end{cases}, \quad (12)$$

where the set Λ_I represents the indices of the visual tokens.

Therefore, the proposed AVA module uses the recurrent state and current visual observation to calculate soft weights to guide the VLA model to filter and focus its attention based on historical information.

3.4. Training and Inference Procedure

The proposed AVA-VLA framework introduces a recurrent dependency through the recurrent state $\boldsymbol{r}^{t-1}$. Training such a recurrent model ideally requires backpropagation through time over the entire trajectory to capture long-term dependencies. However, given the substantial memory constraint and computational cost of modern VLA backbones, performing the full backpropagation through time is computationally prohibitive [34].

To address this challenge, we adopt a truncated backpropagation through time strategy [25]. We unroll the model for a fixed, short horizon. Specifically, for the n-th sample in the training batch, it contains a continuous observation sequence $\{\boldsymbol{x}^{t,n}\}_{t=0}^{T-1}$. For each timestep t in this sequence, we calculate the action chunk prediction loss using the Mean Absolute Error (MAE): $\mathcal{L}^{t,n} = \mathcal{L}(\mathcal{A}^{t,n}, \mathcal{A}_{\mathrm{GT}}^{t,n})$. To prevent overly dispersed soft attention weights, we add an L2 penalty regularizer $\mathcal{L}_{\omega}^{t,n}$ on the mean value of the weight vector $\boldsymbol{\omega}^{t,n}$, defined as:

$$\mathcal{L}_{\omega}^{t,n} = \|\mu(\boldsymbol{\omega}^{t,n}) - c\|, \quad (13)$$

where $\mu(\cdot)$ is the mean function and c is a target mean hyperparameter. This encourages the AVA module to focus on task-relevant regions while suppressing distracting background responses (*see Appendix D for more analysis*).

Therefore, the total loss of one training batch is the sum of the prediction loss and penalty loss of N truncated sequences:

$$\mathcal{L}_{\text{total}} = \sum_{n=1}^{N}\sum_{t=0}^{T-1}(\mathcal{L}^{t,n} + \lambda\mathcal{L}_{\omega}^{t,n}), \quad (14)$$

where N is the batch size, and λ is a balancing coefficient. In our experiments, we set $T=4$ to balance computational feasibility with the need to learn the temporal dynamics captured by the recurrent state. At the first timestep ($t=0$) of any sequence, the initial recurrent state $\boldsymbol{r}^{-1}$ is initialized as a zero embedding, i.e., $\boldsymbol{r}^{-1} = \bar{\mathbf{0}}$.

During inference, the model operates in a fully recurrent manner. At the beginning of a new episode ($t=0$), the initial recurrent state $\boldsymbol{r}^{-1}$ is initialized as the zero embedding. Then, for each subsequent timestep $t \ge 0$, the agent receives the current observation $\boldsymbol{x}^t$ and performs a single forward pass as defined in Eq. (6), conditioned on both $\boldsymbol{x}^t$ and the previously computed recurrent state $\boldsymbol{r}^{t-1}$. This forward pass predicts the action chunk $\mathcal{A}^t$ and simultaneously extracts the recurrent state $\boldsymbol{r}^t$. This loop continues for the entire inference process.

Remark. We explicitly note that the soft weights vector $\boldsymbol{\omega}^t$ computed by the AVA module has a natural application in visual token reduction [24, 54]. Visual tokens with low importance scores can be pruned to reduce the computational cost of the LLM backbone. While this is a valid direction for improving model efficiency, it is not the primary focus of this work. We provide a preliminary exploratory analysis on leveraging the weight vector to do token reduction, which further validates the effectiveness of our proposed method. Details can be found in Section 4.4.

4. Experiments

We evaluate the effectiveness of our approach through a set of experiments spanning both simulation benchmarks and real-world robot manipulation tasks. Additionally, we conduct a comprehensive ablation study and analysis to validate the effectiveness of our approach. All experiments are conducted on Nvidia A800 GPUs.

4.1. Experimental Setup

We conduct experiments on three challenging settings: the LIBERO [28] and CALVIN [31] benchmarks for evaluation in simulation environments, and a real-world table-mounted Mobile ALOHA robot with four test tasks, to validate the sim-to-real transferability of our method. We use the open-source OpenVLA-OFT [20] as our foundation model, which consists of a two-branch vision encoder (DINOv2 and SigLIP) and a LLaMA2-7B backbone [51]. *Due to space limitations, implementation details are provided in Appendix A.*

LIBERO. LIBERO [28] is a benchmark for lifelong robot learning. It uses a Franka Emika Panda arm in MuJoCo, with datasets split into four suites: LIBERO-Spatial, LIBERO-Object, LIBERO-Goal, and LIBERO-Long. It contains 5,000 episodes across 100 tasks. Data includes RGB images, proprioceptive states, and delta actions, with procedural generation for diversity. *LIBERO+ [11] is a challenging LIBERO-based benchmark, which offers a robust benchmarking framework with 7 perturbation dimensions and 21 sub-dimensions. It allows users to assess model performance across various challenges systemati-*

Table 1. Comparison on the LIBERO benchmark. The results are reported in two groups: one policy for all 4 suites, and one policy per suite. The best results in each column of each group are highlighted in **bold**.

Method	Spatial SR (%)	Object SR (%)	Goal SR (%)	Long SR (%)	Average SR (%)
One policy for all 4 suites					
TraceVLA [64]	84.6	85.2	75.1	54.1	74.8
WorldVLA [6]	87.6	96.2	83.4	60.0	81.8
π_0 [3]	96.8	98.8	95.8	85.2	94.2
π_0-FAST [36]	96.4	96.8	88.6	60.2	85.5
UnifiedVLA [56]	95.4	98.8	93.6	94.0	95.5
OpenVLA-OFT [20]	**97.7**	98.0	96.1	95.3	96.8
AVA-VLA (Ours)	97.4	**99.4**	**97.4**	**97.6**	**98.0**
One policy per suite					
OpenVLA [19]	84.7	88.4	79.2	53.7	76.5
SpatialVLA [39]	88.2	89.9	78.6	55.5	78.1
CoT-VLA [63]	87.5	91.6	87.6	69.0	83.9
NORA [15]	92.2	95.4	89.4	74.6	87.9
PD-VLA [45]	95.5	96.7	94.9	91.7	94.7
UniVLA [5]	96.5	96.8	95.6	92.0	95.2
OpenVLA-OFT [20]	97.6	98.4	97.9	94.5	97.1
FLOWER [41]	97.5	99.1	96.1	94.9	96.9
RIPT-VLA [48]	99.0	98.6	**98.6**	93.8	97.5
AVA-VLA (Ours)	**99.2**	**99.6**	97.9	**96.2**	**98.2**

cally. We conduct additional experiments on the LIBERO+ benchmark, and the results are put in Appendix D.

CALVIN. CALVIN [31] is a simulated benchmark for language-conditioned, long-horizon manipulation, using a Franka Panda arm with RGBD observations, proprioception, and natural language goals. It evaluates sequential reasoning in VLA. CALVIN spans 34 tasks across four environments (A-D), with 20,000+ episodes, emphasizing unseen object generalization and multi-stage sequences (e.g., "open drawer, pick blue block, push into drawer"). Following [5], we used the CALVIN "ABC→D" setting, which means training on environments A, B, and C and evaluating on environment D, to evaluate performance on the zero-shot generalization tasks.

Mobile ALOHA Real-Robot Experiments. We use a stationary cobot magic dual-arm robot to assess our model's adaptability to novel real-world environments with a small number of robot demonstrations. Following [5, 63], we perform evaluations across four challenging tasks. These include Pick and Place, which involves placing the bucket in the center, and then placing irregular-shaped objects into the bucket (e.g., "put <obj> into bucket"), and Sequenced Instruction Understanding, which requires executing multi-step commands like stacking a Tower of Hanoi ("Stack tower of hanoi"). We also test Flexible Object Folding, a deformable object manipulation task requiring a specific three-stage process to fold a towel ("fold towel twice"), and Dexterous Action, which involves fine-motor skills such as using a shovel to scoop small items (e.g., corn, sesame seeds) into a bowl. For each task, the dataset contains between 30 and 450 demonstrations. *Details of the task suites are provided in Appendix B.*

Table 2. Comparison on the CALVIN ABC→D benchmark. The results are reported in terms of success rates (%) and average length. The best results in each column are highlighted in **bold**.

CALVIN ABC→D	Task completed in a row ↑					Avg. len ↑
	1	2	3	4	5	
OpenVLA [19]	91.3	77.8	62.0	52.1	43.5	3.27
UniVLA [5]	95.5	85.8	75.4	66.9	56.5	3.80
UnifiedVLA [56]	98.9	94.8	89.0	82.8	75.1	4.41
OpenVLA-OFT [20]	96.9	92.0	85.7	80.4	72.9	4.28
FLOWER [41]	99.4	95.8	90.7	84.9	77.8	4.53
VLA-Adapter [55]	99.1	94.6	88.8	82.8	76.5	4.42
Seer [50]	96.3	91.6	86.1	80.3	74.0	4.28
AVA-VLA (Ours)	**99.6**	**97.6**	**94.1**	**89.9**	**84.1**	**4.65**

4.2. Evaluation Results

Baselines. We selected recently published works' main method as baselines. They are TraceVLA [64], WorldVLA [6], π_0 [3], π_0-FAST [36], UnifiedVLA [56], OpenVLA-OFT [20], OpenVLA [19], SpatialVLA [39], CoT-VLA [63], NORA [15], PD-VLA [45], UniVLA [5], OpenVLA-OFT [20], FLOWER [41], RIPT-VLA [48], VLA-Adapter [55], Seer [50]. The results of these baselines in LIBERO and CALVIN benchmarks are based on original references or other published works, ensuring objectivity and correctness. For Mobile ALOHA real-robot experiments, we select UniVLA and OpenVLA-OFT methods as baselines.

Evaluation Metrics. We use widely adopted performance evaluation metrics "Success Rate (SR)" (the same

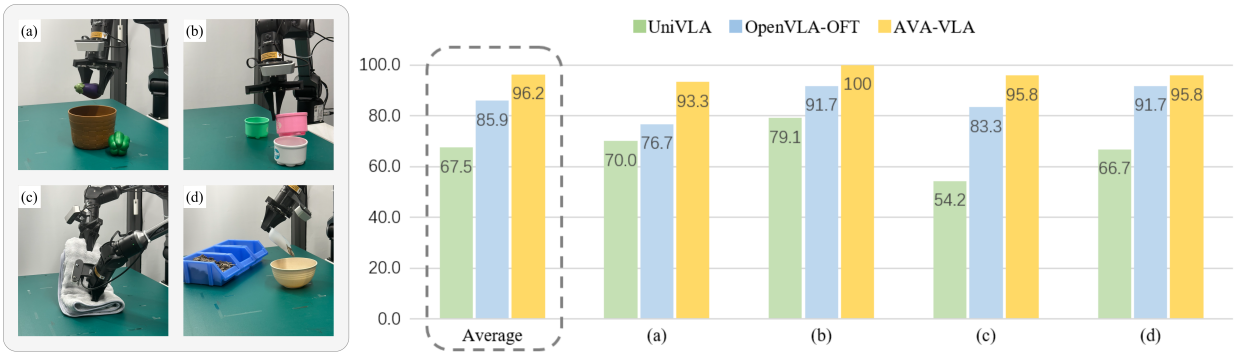


Figure 3. Comparison on the Mobile ALOHA real-world experiments. Evaluation across four manipulation tasks, including (a) Pick and Place, (b) Sequenced Instruction Understanding, (c) Flexible Object Folding, (d) Dexterous Action. Left: Representative middle states for each task setup. Right: Task-specific success rates and cross-task averages for our method and baselines.

in LIBERO [28]) to evaluate the results for three challenging settings. In addition, we use "Average len" of completed tasks (the larger the better, with values between 0-5) as metrics for the CALVIN benchmark.

LIBERO. We present quantitative results in Table 1. Following established baselines, we conduct experiments in two different settings: training policies on each task suite independently (single-task learning) and training a single policy for all task suites (multi-task learning [26]). Results demonstrate that the proposed AVA-VLA framework achieves state-of-the-art overall performance in both single-task and multi-task settings. Moreover, it consistently achieves the best performance on the most challenging LIBERO-Long task suite. These results demonstrate the superiority of the proposed AVA-VLA framework.

CALVIN. We present the success rates for each task and the average completed length across all five tasks of the CALVIN benchmark in Table 2. The results show that the proposed AVA-VLA framework comprehensively outperforms baseline methods across all tasks. This demonstrates our method's strong generalization ability, with an average length superior to previous state-of-the-art baselines.

Mobile ALOHA. The Pick and Place task is evaluated for a total of 30 trials (10 per object), while other tasks are evaluated for 24 trials each. The experimental results on real-world tasks are reported in Figure 3. In this setting, models are fine-tuned on a relatively small set of demonstrations. The results demonstrate that the proposed model possesses robust semantic understanding and dexterous action capabilities after training. Overall, AVA-VLA achieves the highest average performance compared to baseline approaches, confirming its real-world applicability. *We visualize the execution trajectories for these tasks in Appendix B.*

4.3. Ablation Studies

Model Backbones. To validate the effectiveness of the proposed framework, following [55], we compare three kinds of backbones: The OpenVLA-7B backbone [19] pre-trained on robotic data, the prismatic VLM trained on LLaMA2-7B [51], and the prismatic VLM [18] trained on Qwen2.5-0.5B [49]. The last two are different-scale backbones without pre-training on robotic data. We compare the proposed method against the standard OpenVLA-OFT method on the LIBERO-Long task suite in the single task setting. Results reported in Table 3 show that our method improves performance across different backbones, even on backbones not pre-trained on robotic datasets.

Table 3. Ablation study on the model backbones. Comparison on the LIBERO-Long task suite in the LIBERO benchmark in terms of success rates (%). The best results of each model backbone setting are highlighted in **bold**.

Backbones	OpenVLA-OFT	AVA-VLA
OpenVLA-7B [19]	94.5	**96.2** (1.7% ↑)
LLaMA2-7B [51]	90.0	**92.6** (2.6% ↑)
Qwen2.5-0.5B [49]	89.4	**90.8** (1.4% ↑)

AVA Module and State-Based Initialization. The AVA-VLA framework consists of two components: the state-based initialization strategy and the AVA module. To validate their individual effectiveness, we conduct ablation experiments on the LIBERO benchmark. As shown in Table 4, the two recurrent-state-driven components are complementary. State-based initialization injects the recurrent state into the action placeholder to preserve temporal context, which is especially beneficial on LIBERO-Long. The AVA module uses the recurrent state to reweight visual tokens and suppress irrelevant visual content, leading to consistent gains across suites. Each component alone improves over OpenVLA-OFT, and their combination achieves the best overall performance.

4.4. Analysis

Qualitative Visualization. To investigate the AVA module's focus during task execution, we visualize the soft weights ω^t across visual tokens during inference. As illus-

Table 4. Ablation study on the two key components in the AVA-VLA framework. The results on LIBERO in terms of success rates (%) under the "one policy for all 4 suites" setting are reported. The best results in each column are highlighted in **bold**.

Method	Spatial SR (%)	Object SR (%)	Goal SR (%)	Long SR (%)	Average SR (%)
OpenVLA-OFT	97.7	98.0	96.1	95.3	96.8
AVA-VLA (State-based initialization)	97.2	98.8	96.6	97.2	97.5
AVA-VLA (AVA module)	**97.8**	98.6	97.0	96.6	97.5
AVA-VLA (AVA module + State-based initialization)	97.4	**99.4**	**97.4**	**97.6**	**98.0**

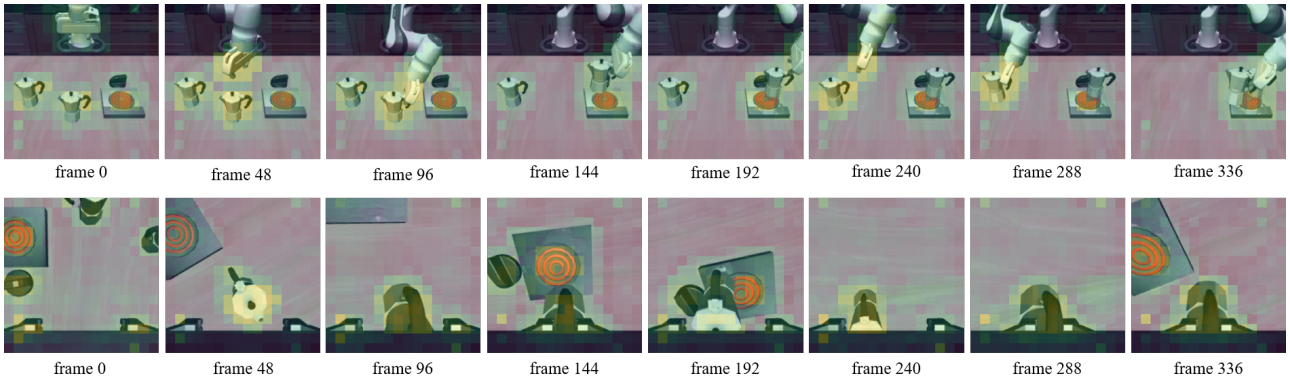


Figure 4. Visual dynamics. The evolution of soft weights during the task "put both moka pots on the stove" from two viewpoints.

Table 5. Study on the visual token pruning with different pruning ratios. The results on LIBERO in terms of success rates (%) under the "one policy for all 4 suites" setting are reported.

Pruning Ratio	Spatial SR (%)	Object SR (%)	Goal SR (%)	Long SR (%)	Avg. SR (%)
0%	97.4	99.4	97.4	97.6	98.0
50%	97.2	99.4	97.2	95.2	97.3
60%	97.6	99.4	97.0	95.0	97.3
70%	97.4	99.2	98.0	94.6	97.3
80%	96.8	98.2	96.2	92.8	96.0
90%	94.2	97.8	94.2	89.2	93.9

trated in Figure 4, the attention weights consistently concentrate on the robotic arm's contact regions and the target objects. This selective focus demonstrates the module's ability to identify task-relevant visual features, validating its effectiveness. Furthermore, a direct comparison in Figure 1 reveals that while the vanilla OpenVLA-OFT baseline fails to localize the task-relevant region across viewpoints, AVA-VLA maintains a robust and spatially consistent focus by effectively leveraging historical context.

Visual Token Reduction. The proposed AVA module has a potential application in visual token reduction. Although visual token pruning causes the model to lose some visual information, it reduces the model's computational cost and is beneficial for efficient inference [17, 52, 56, 58]. To validate that our AVA module effectively prioritizes task-relevant information, we apply a direct ranking strategy to prune visual tokens during inference. Specifically, for a given pruning ratio, we rank all visual tokens by their soft weights and retain only the top-ranked portion corresponding to the desired retention percentage. The results reported in Table 5, demonstrate the robustness of our method: the model suffers only a negligible drop in performance after pruning. Notably, with pruning ratios of 50%, 60%, and 70%, the proposed method continues to outperform the OpenVLA-OFT and maintains performance comparable to the state-of-the-art baselines reported in Table 1. The decline in success rate mainly comes from the most challenging LIBERO Long task suite, while the results remain consistent across the other task suites. Even after reducing 90% of the visual tokens, our method still outperforms many baseline methods listed in Table 1. This result further demonstrates the effectiveness of the proposed AVA-VLA framework.

5. Conclusion

In this paper, we reformulate robot manipulation from a POMDP perspective and propose AVA-VLA, a novel vision-language-action framework for temporally grounded decision-making. Unlike prior VLA models that process each frame independently, our method introduces a recurrent state to approximate the agent's belief, and builds an Active Visual Attention module to dynamically modulate visual processing of the current observation. In this way, AVA-VLA can actively suppress irrelevant information and focus on task-critical visual features based on historical context. Extensive experiments demonstrate the superiority of AVA-VLA, achieving state-of-the-art performance across multiple robot simulation benchmarks, including LIBERO and CALVIN, and transferring effectively to diverse real-world robotic tasks. These results highlight the value of recurrent state modeling and history-aware visual processing for robotic sequential decision-making.

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